Naïve Bayes

↳ Supervised ML algorithm used for classification problem.

↳ Here, we calculate the probability.

★ Bayes Theorem :-

Let two events A & B

Acc. to conditional probability :

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

or $P(A \cap B) = P(A|B) \cdot P(B)$ ——— (1)

Similarly,

$$P(B|A) = \frac{P(A \cap B)}{P(A)}$$

or $P(A \cap B) = P(B|A) \cdot P(A)$ ——— (2)

On comparing eq. (1) & (2) [LHS is same, so RHS will also be same]

$$P(A|B) \cdot P(B) = P(B|A) \cdot P(A)$$

or

$$\boxed{P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}}$$

This is Bayes Theorem.

or

$$\text{Posterior} = \frac{\text{Likelihood} \times \text{Prior}}{\text{Evidence}}$$

Ex- Suppose A = Rain, B = Clouds

So,

$$P(\text{Rain} \mid \text{Clouds}) = \frac{P(\text{Clouds} \mid \text{Rain}) \times P(\text{Rain})}{P(\text{Clouds})}$$

Studying Bayes Theorem using Naive Bayes :-

Ex- Email Classification [spam → 1 , not spam → 0]

Assume a piece of text & we have to identify whether the text belongs to class spam or not spam.

Text = "You won a Jackpot"

Here, class probability belongs to y=0 or y=1

$P(y=0 \mid \text{text})$ or $P(y=1 \mid \text{text})$

∴ Using Bayes Theorem →

(Text = t)

$$P(y=0 \mid t) = \frac{P(t \mid y=0) \cdot P(y=0)}{P(t)} \quad \text{—— (1)}$$

$$P(y=1 \mid t) = \frac{P(t \mid y=1) \cdot P(y=1)}{P(t)} \quad \text{—— (2)}$$

discard this term as it does not add any difference

$P(t)$ ——————————— does not add any difference in the O/P

∴ equations will become →

$$P(y=0|t) = P(t|y=0) \cdot P(y=0)$$
$$P(y=1|t) = P(t|y=1) \cdot P(y=1)$$

→ Individual probabilities
Easier to calculate

$P(t|y=0)$ and $P(t|y=1)$ ↓ difficult to calculate

Suppose text t consists of words w_1, w_2, w_3 -----

So, equations will become →

$$P(y=0|t) = P(w_1 \cap w_2 \cap w_3 \cap w_4 \text{ -----} | y=0) \cdot P(y=0) \quad \text{——— (1)}$$
$$P(y=1|t) = P(w_1 \cap w_2 \cap w_3 \cap w_4 \text{ ----} | y=1) \cdot P(y=1) \quad \text{——— (2)}$$

[$w_1 \cap w_2 \cap w_3$ ---- represents the occurrence of words together]

In the above equations → $P(w_1 \cap w_2 \cap w_3 \text{ ---} | y=0)$ & $P(w_1 \cap w_2 \cap w_3 \text{ ---} | y=1)$

↓

To simplify this term, we take an assumption

↙

All the words are independent of each other

↓

This assumption is known as

"Naive Assumption"

so,

$$P(w_1 \cap w_2 \cap w_3 \cdots | y=0) = P(w_1|y=0) \cdot P(w_2|y=0) \cdot P(w_3|y=0) \cdots$$

$$P(w_1 \cap w_2 \cap w_3 \cdots | y=1) = P(w_1|y=1) \cdot P(w_2|y=1) \cdot P(w_3|y=1) \cdots$$

∴ final formula →

$$P(y=0|t) = P(w_1|y=0) \cdot P(w_2|y=0) \cdot P(w_3|y=0) \cdots \cdot P(y=0)$$

$$P(y=1|t) = P(w_1|y=1) \cdot P(w_2|y=1) \cdot P(w_3|y=1) \cdots \cdot P(y=1)$$

or

$$P(y=0|t) = P(y=0) \cdot \prod_{i=1}^{n} P(w_i|y=0)$$

$$P(y=1|t) = P(y=1) \cdot \prod_{i=1}^{n} P(w_i|y=1)$$

The above formula is the formula for Naive Bayes Algorithm.

As we all know that the range of probability is b/w 0 to 1

so, if we multiply many probabilities (many small nos.), the result will be a very small no.

$$0.6 \times 0.3 \times 0.25 \times 0.5 \times 0.28 \times 0.7 = 0.00441$$

↓

This is known as the problem of underflow

↳ Computer may round it off to zero

↙

This causes the problem of numerical instability in Naive Bayes

∴ To avoid this issue, change multiplication ⟶ addition

How? → take log

{ Identity in logarithm →

$\log(m * n) = \log m + \log n$

So, the multiplication term will become →

$= \log\left[P(w_1|y=0) \cdot P(w_2|y=0) \cdots\right]$

($P(w_1|y=0)$ = m, $P(w_2|y=0)$ = n)

$= \log\left[P(w_1|y=0)\right] + \log\left[P(w_2|y=0)\right] + \log\left[P(w_3|y=0)\right] + \cdots$

↳ This reduces the prob. of numerical instability

∴ Acc. to Naive Bayes →

$$P(t|y=0) = P(y=0) \cdot \sum_{i=1}^{n} \log [P(w_i|y=0)]$$

$$P(t|y=1) = P(y=1) \cdot \sum_{i=1}^{n} \log [P(w_i|y=1)]$$